

EDS 223 Assignment 1

Map making practice

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Air Quality in Oahu, Hawaii via EJScreen Tool

The two maps created focus on air quality and environmental injustice in Oahu, Hawaii, specifically, in the lower regions where there are larger tourist attractions and infrastructures. Though, Hawaii has a long history of environmental injustice due to many factors outside this scope, the two maps created below show the air quality in a heavily populated area and a higher risk of air cancer, despite Hawaii having clean air overall.

The first map displays the percentage of lifetime risk of cancer, as listed by the EPA's Toxics Release Inventory (TRI) Program, which is mostly concentrated at the bottom of Oahu, as mentioned above. In the EJScreen technical documentaion (2023), the term "cancer" in the dataset is defined as the "lifetime cancer risk from inhalation of toxics". The second map focuses on the toxicity concentration from the RSEI model that was found in the air. The raw values range from 0 to 4,512,172.1, with Oahu having the max at 4,512. Thus, map 1 shows the percentile of high risk areas in danger of cancer from toxic chemicals, which was calculated by persons per million over a 70-year lifetime, with the max in Oahu being 50 per million. In simpler terms, there would be an estimated 50 additional cancer cases per million over a 70-year lifetime according to EPA. For map 2, percentiles are used, " with a median at the 50th percentile, which corresponds to the raw value of 603.032"(EJScreen, 2023). In the case for Oahu, the highest percentile is roughly at the 78th percentile or a raw value at 1,078.

Those living or working near these areas are the ones consistently breathing the air, which puts them in the risk of inhaling toxic chemicals and/or developing air cancer. This can impact those who face poverty, and have no means to relocate or choose a different area for work. This can be further explored by creating a map with "low income" to show those living in the vicinity. Lastly, it's important to note that map 1 and map 2 have a strong correlation with the location of two major infrastructures, besides the tourist attractions, on the lower region of the island near Honolulu. These two are the Daniel K. Inouye International Airport and Par Hawaii Refining, an energy plant, that had a recent settlement in 2022 due to Clean Air Violations with the EPA. As mentioned, these two areas are also major tourist attractions,

with a strip of hotels and multiple shopping areas that can be the cause of increased amount of anthropogenic pollution emitted on the island of Oahu.

Load packages

```
library(tidyverse) # For R
library(sf) # Read vector
library(here) # Locate files
library(stars)
library(tmap) # For static and interactive maps
library(janitor) # Clean data
```

Read EJScreen and Data

```
# read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data",
  "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"),
  quiet = TRUE)

# filter to a state you are interested in
hawaii <- ejscreen |>
  dplyr::filter(ST_ABBREV == "HI")

# filter to a county you are interested in
honolulu <- ejscreen |>
  dplyr::filter(CNTY_NAME %in% c("Honolulu County"))

# find the average values for all variables within counties
hawaii_counties <- aggregate(hawaii,
  by = list(hawaii$CNTY_NAME), FUN = mean)
```

Data Cleaning

```
# Convert all columns to lower case
hawaii <- hawaii |>
  clean_names()
```

```

honolulu <- honolulu |>
  clean_names()

hawaii_counties <- hawaii_counties |>
  clean_names()

```

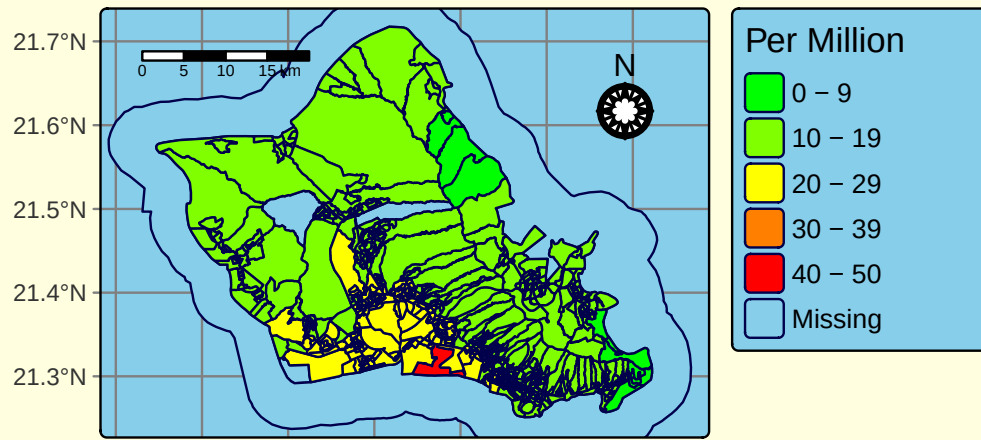
Map 1: Percentage of Lifetime Risk of Cancer from Inhaling Toxic Air

```

# Select primary spatial object as county of choice
m1 <- tm_shape(honolulu,
bbox = c(-158.3042, 21.2367, -157.6251, 21.7293)) +
  tm_graticules() +
  tm_polygons(fill = "cancer", # Map percentile of air cancer
    fill.legend = tm_legend(title = "Per Million", # Create legend
    bg.color = "skyblue",
    width = 8),
    fill.scale = tm_scale_intervals(
      values = c("green", "yellow", "red"),
      value.na = "skyblue")) +
  tm_compass( # Customize compass
    type = "rose",
    size = 1.5,
    position = c(
      "right", "top")) +
  tm_scalebar(position = c("left", "top"),
    text.size = .4) +
  tm_title(
    text = "Lifetime Cancer Risk from
    Toxic Air in Oahu Cases Per Million (EJScreen, 2019)",
    size = 1) + # Add title to map
  tm_layout(scale = 1.2,
    asp = 0, # Customize backgrounds
    style= "beaver",
    bg.color = "skyblue",
    outer.bg.color = "lightyellow")
m1

```

Lifetime Cancer Risk from
Toxic Air in Oahu Cases Per Million (EJScreen, 2019)



Map 2: Toxic Air Chemicals

```
m2 <- tm_shape(honolulu,
bbox = c(-158.3042, 21.2367, -157.6251, 21.7293)) +
  tm_graticules() +
  tm_polygons(fill = "p_rsei_air",
    fill.legend = tm_legend(title = "Percentile",
      bg.color = "skyblue",
      width = 8),
    fill.scale = tm_scale_intervals(
      values = c("green", "yellow", "red"),
      value.na = "skyblue")) +
  tm_compass(
    type = "rose",
    size = 1.5,
    position = c("right", "top")) +
  tm_scalebar(position = c("left", "top"),
    text.size = .4) +
  tm_title(text = "Percentile of Toxic Chemicals in Oahu (EJScreen 2019)",
    size = 1) +
  tm_layout(scale= 1.2,
```

```

asp = 0,
style = "beaver",
bg.color = "skyblue",
outer.bg.color = "lightyellow")

```

m2

